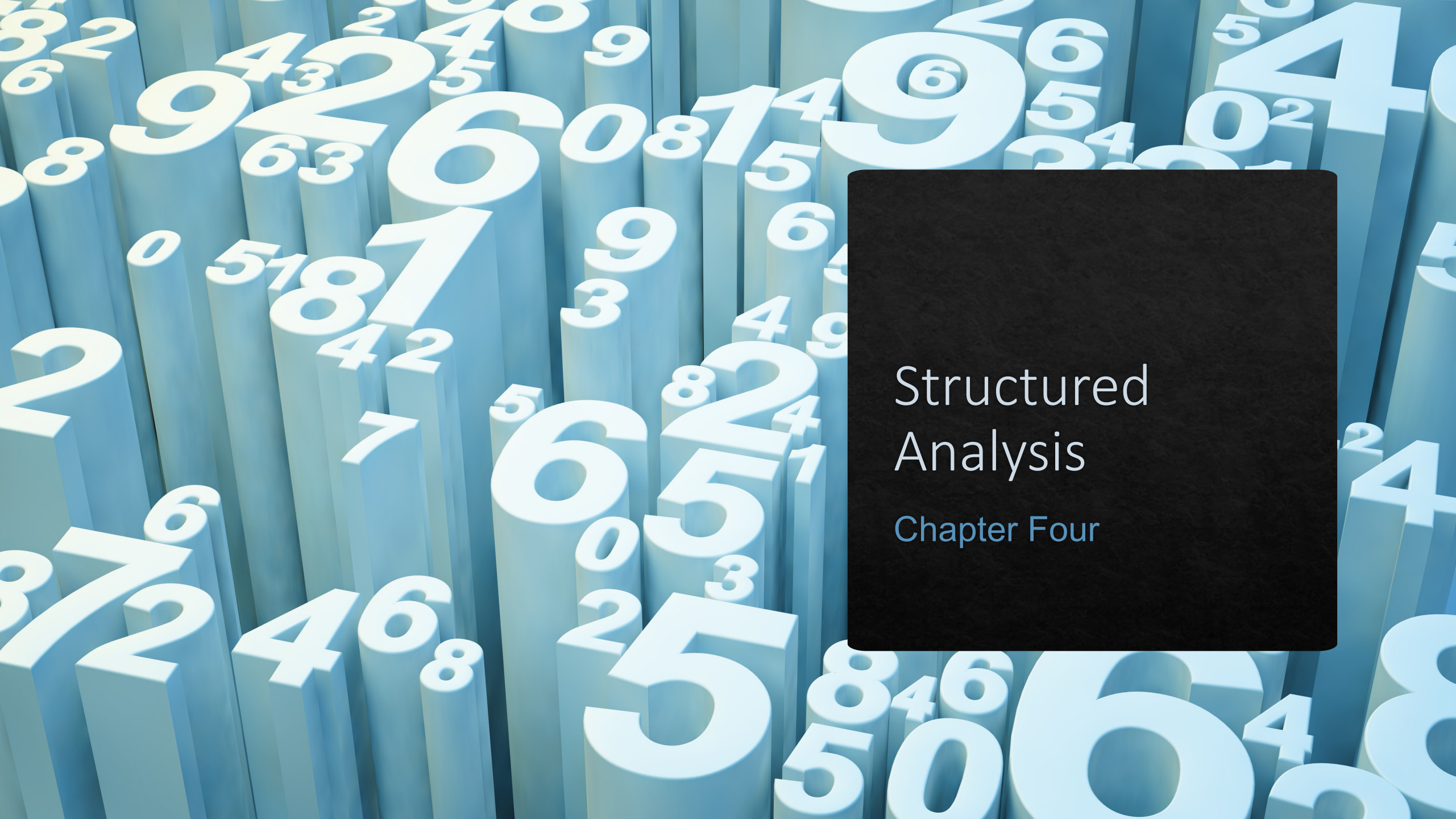


The background of the slide is a dense, 3D-rendered field of numbers. The numbers are in various shades of light blue and white, creating a sense of depth and texture. They are scattered across the entire frame, with some numbers appearing larger and more prominent than others. The lighting is soft, highlighting the edges of the numbers and creating subtle shadows.

Structured Analysis

Chapter Four

Structured Analysis

- ◆ Let us see few analysis and design tools used by software designers:
 - ◆ Data Flow Diagrams
 - ◆ Data Dictionary
 - ◆ Decision Trees
 - ◆ Decision Tables

Data flow Diagram

- ◆ DFD is graphical representation/model of flow of data in an information system.
- ◆ It is capable of depicting incoming data flow, outgoing data flow and stored data.
- ◆ Data Flow Diagrams types.
 - ◆ **Logical DFD -**
 - ◆ It concentrates on the system process, and flow of data in the system.
 - ◆ **Physical DFD -**
 - ◆ shows how the data flow is actually implemented in the system.
 - ◆ It is more specific and close to the implementation.

Data flow diagram

□ **DFD Components**

◇ **Entities -**

- ◇ Entities are source and destination of information data.
- ◇ Entities are represented by a rectangles with their respective names.

◇ **Process -**

- ◇ Activities and action taken on the data are represented by Circle or Round-edged rectangles.

◇ **Data Storage -**

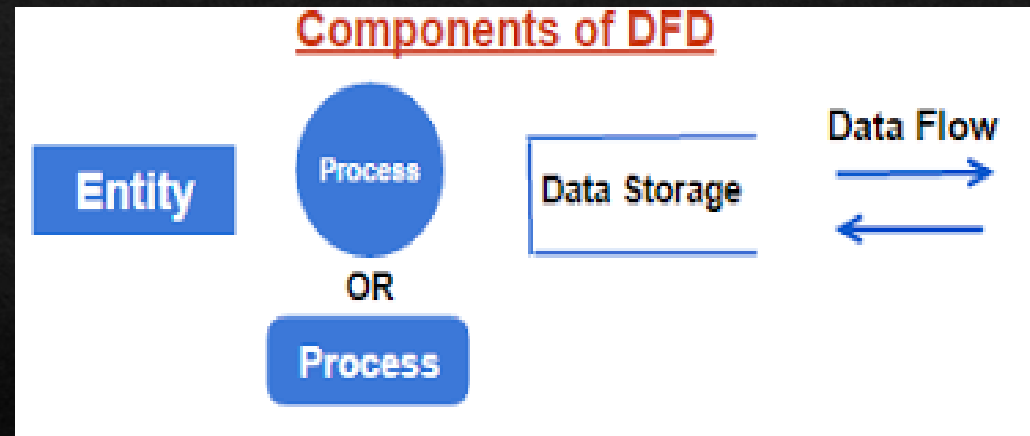
- ◇ A location where data is saved to or retrieved from , such as a data base

◇ **Data Flow -**

- ◇ Movement of data is shown by pointed arrows.
- ◇ Data movement is shown from the base of arrow as its source towards head of the arrow as destination.

Data flow diagram

◆ DFD Components : Graphical representation



Data flow diagram

- ◆ **Levels of DFD**

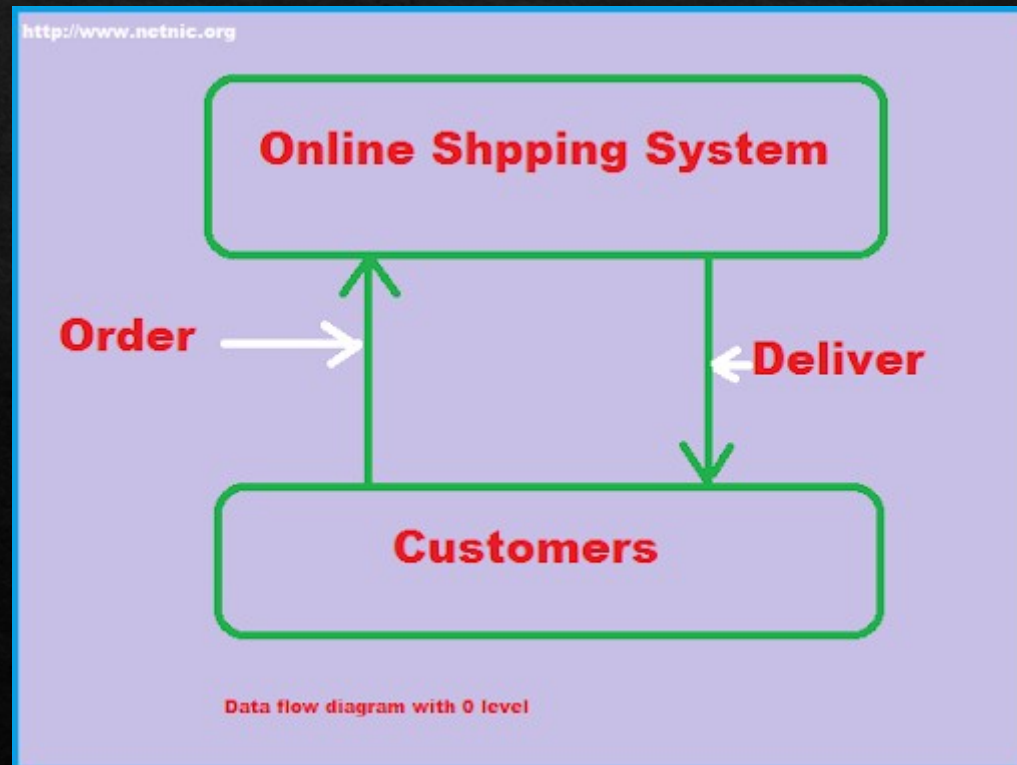
- ◆ **Level 0 -**

- ◆ it is the highest abstraction level which depicts the entire information system as one diagram hiding all the underlying details.
 - ◆ they are also known as context level DFDs.

Data flow diagram

◆ Levels of DFD

- ◆ level 0 can be as easy as the following ecommerce App diagram :



Data flow diagram

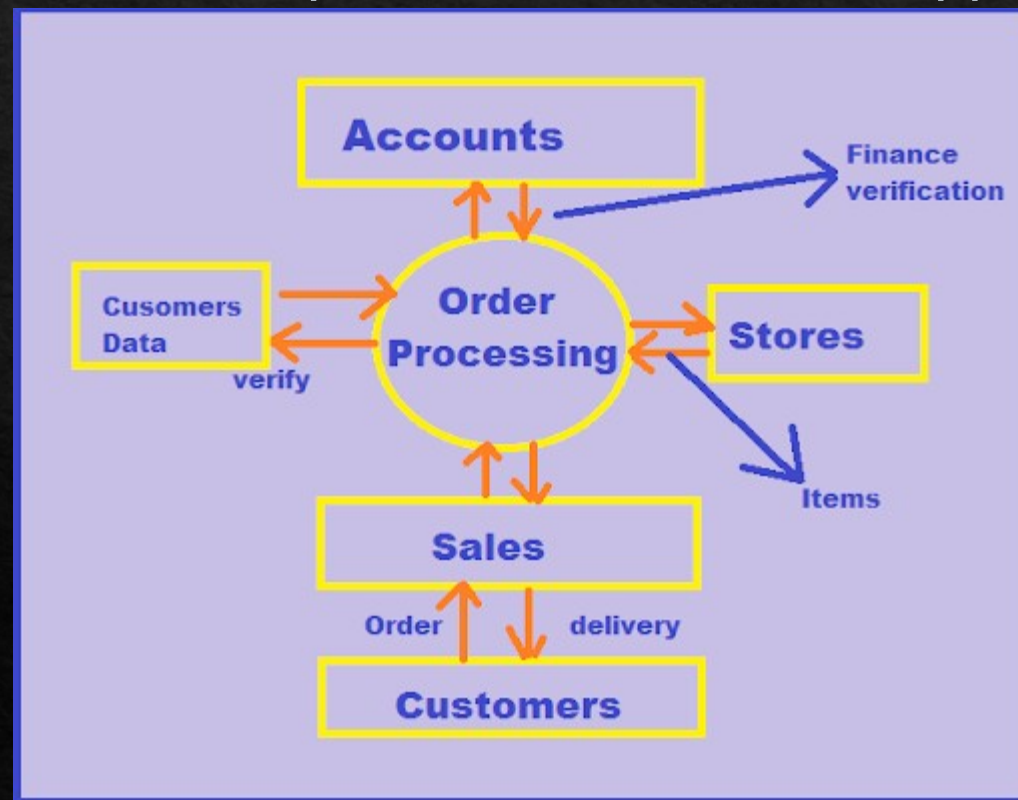
- ◆ **Levels of DFD**

- ◆ **Level 1 -**

- ◆ The Level 0 DFD is broken down into more specific, Level 1 DFD.
 - ◆ Level 1 DFD shows basic modules in the system and flow of data among various modules.
 - ◆ It also mentions basic processes and sources of information.

Data flow diagram

- ◆ Level 1 illustration for the previous ecommerce App.



Data flow diagram

- ◆ **Levels of DFD**

- ◆ **Level 2 -**

- ◆ At this level, DFD shows how data flows inside the modules mentioned in Level 1.

- ◆ Higher level DFDs can be transformed into more specific lower level DFDs

- ◆ with deeper level of understanding unless the desired level of specification is achieved.

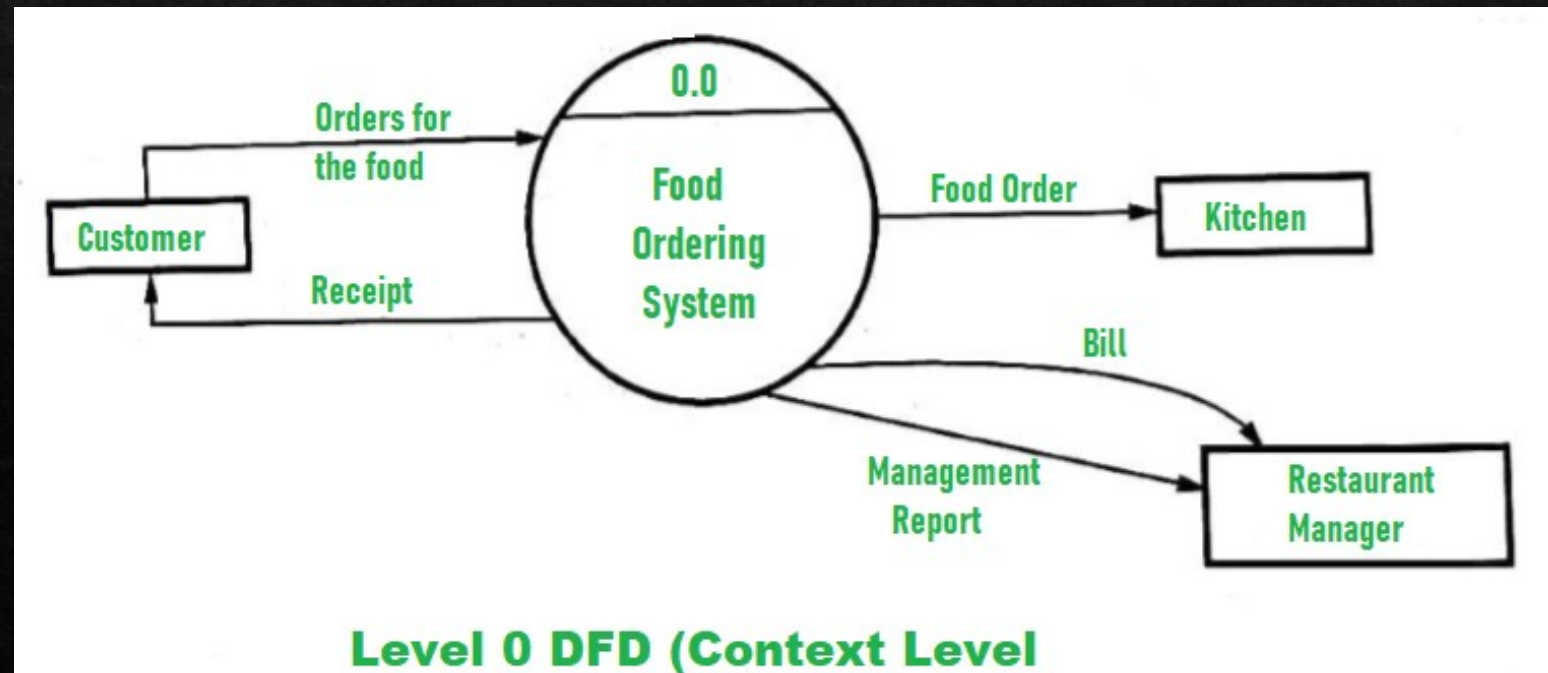
Example

- ◆ Assume you have Food Ordering System

- ◆ **Level 0 DFD**

- ◆ At this level, the Input and Output of the system are shown.

- ◆ The system is designed and established across the world with input and output at this level.



Example

◇ Assume you have Food Ordering System

◇ **Level 0 DFD**

◇ Food Ordering System has the following **input** :

- Food order is input as the customer's order for food.

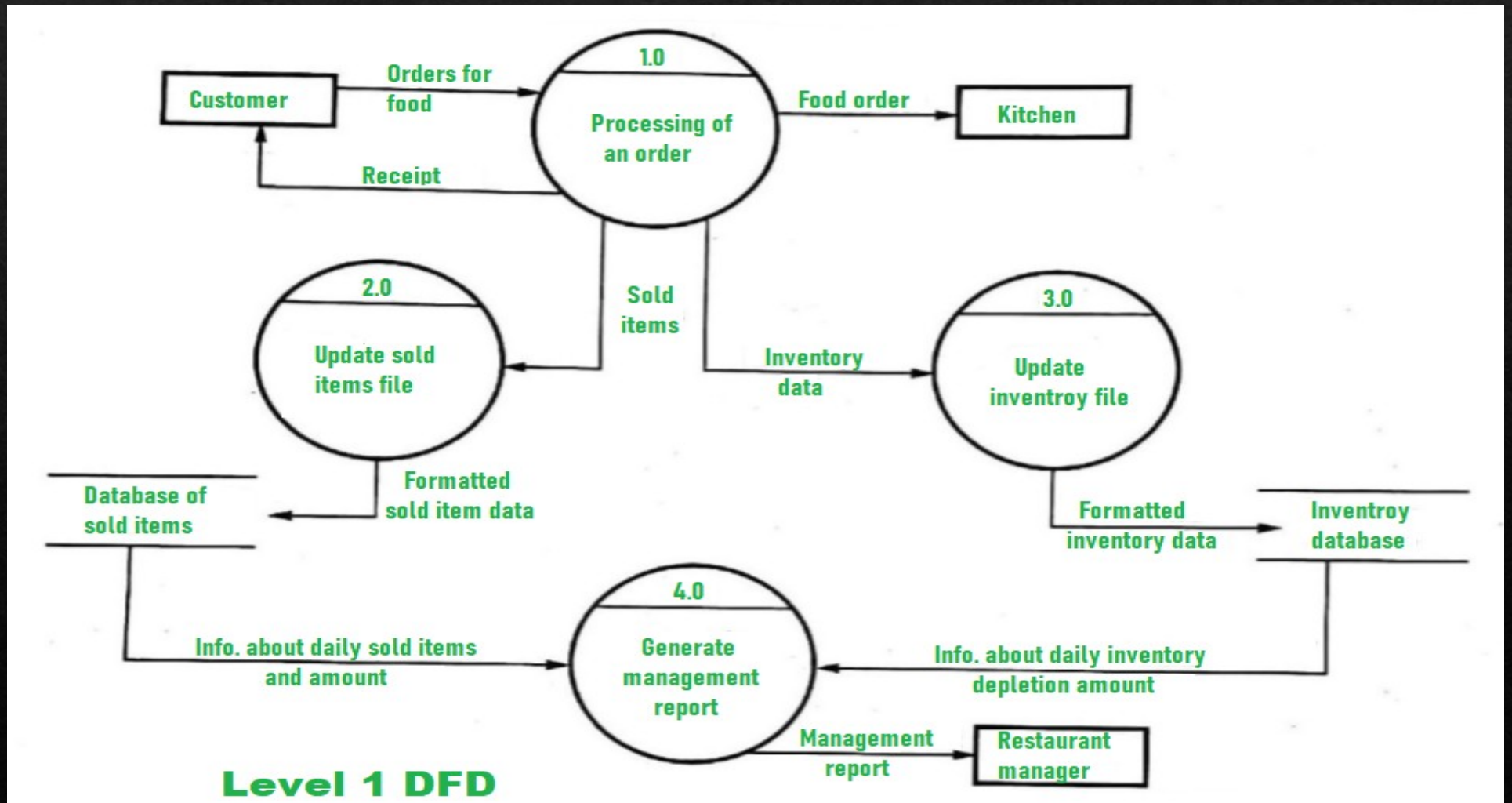
◇ Food Ordering System has the following **output**:

- Receipt of the order.
- For further processing the order, the food order is passed to the kitchen
- The restaurant manager gets the report of Bill and Management

Example

◇ Assume you have Food Ordering System

◇ Level 1 DFD –

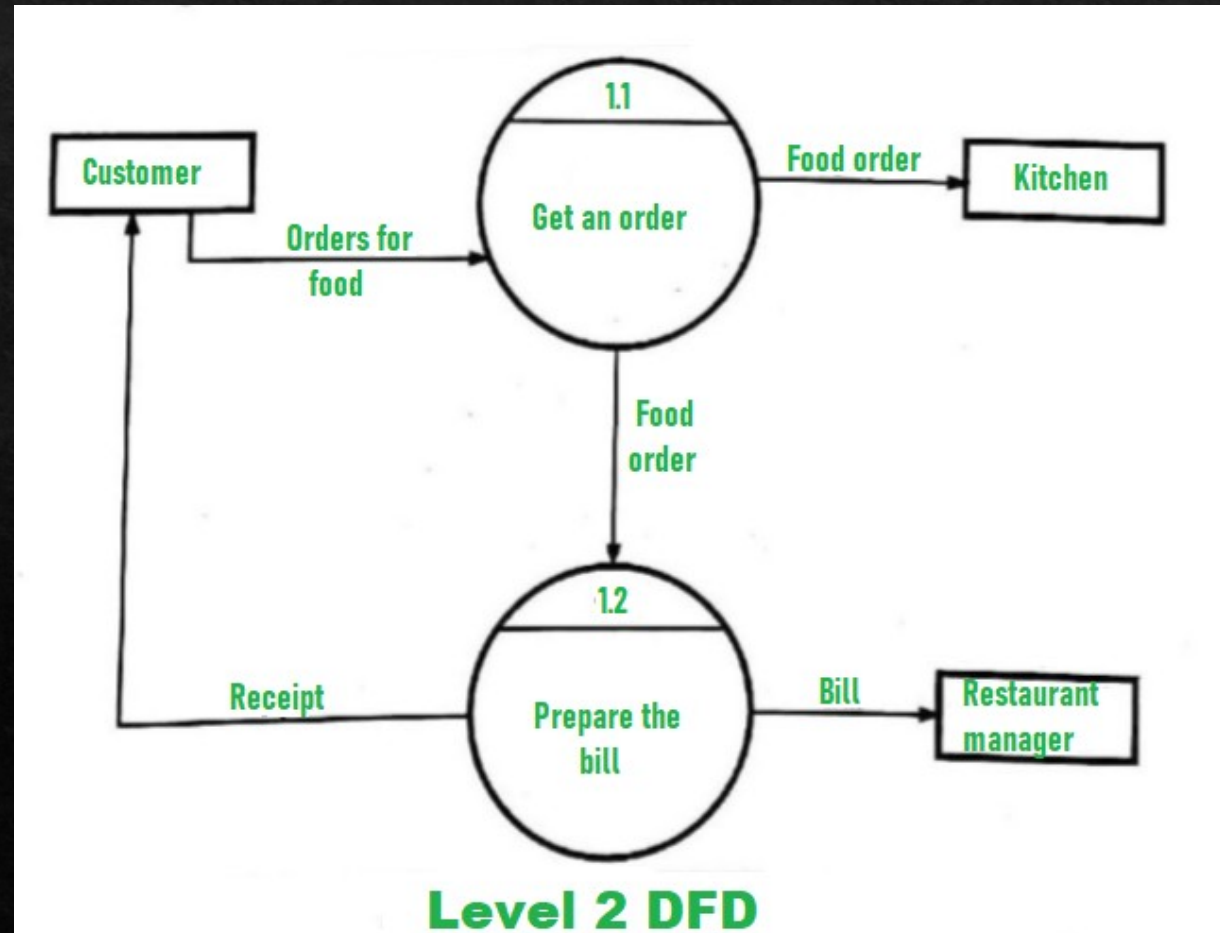


Example

- ◆ Assume you have Food Ordering System
 - ◆ **Level 1 DFD –**
 - ◆ For processing the order, process 1.0 is responsible.
 - ◆ For food, the housekeeping activities involved are represented by processes 2.0, 3.0, and 4.0.
 - ◆ The detailed information about **daily sold items** should be available to create and report management and
 - ◆ The **list of items** that are available '**in-stock**' should be kept by maintaining the **inventory data** (describes the records of datasets such as their name, their content, source, many useful information, etc.) at the same time.
 - ◆ Hence, two data stores are used in this level of DFD which are given below :
 - Database of Sold items
 - Inventory database

Example

- ◆ Assume you have Food Ordering System
- ◆ **Level 2 DFD –**
- ◆ Detailed information about “Processing of an Order” is shown below



Data dictionary

- ◊ A data dictionary is a structured repository of data elements in the system.
- ◊ It stores the descriptions of all DFD data elements that is,
 - ◊ details and definitions of **data flows**, **data stores**, **data** stored in data stores and the **processes**.
- ◊ A data dictionary improves the communication between the analyst and the user

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data dictionaries contain:

- ◆ Data flow.

- ◆ Data structures.

- ◆ Elements.

- ◆ Data stores

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Flow

- ◆ Each data flow should be defined with descriptive information and its composite structure or elements.

- ◆ It Includes the following information

- ◆ ID - identification number, Label, the text that should appear on the diagram

- ◆ A general description of the data flow.

- ◆ The source of the data flow

- ◆ This could be an external **entity**,

- ◆ a **process**, or a data flow coming from a **data store**.

- ◆ The destination of the data flow

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Flow

- ◆ Data flow Include the following information

- ◆ Type of data flow, either:

- ◆ A record entering or leaving a file.

- ◆ Containing a report, form, or screen or

- ◆ Internal used between processes.

- ◆ The name of the data structure or elements

- ◆ The volume per unit time

- ◆ This could be records per day or any other unit of time.

- ◆ An area for further comments and notations about the data flow

Data dictionary

- ◇ Data Dictionary Contents

- ◇ Data Flow

- ◇ Example

Name	Customer Order
Description	Contains customer order information and is used to update the customer master and item files and to produce an order record.
Source	Customer External Entity
Destination	Process 1, Add Customer Order
Type	Screen
Data Structure	Order Information
Volume/Time	10/hour
Comments	An order record contains information for one customer order. The order may be received by mail, fax, or by telephone.

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Structures

- ◆ Data structures are a group of smaller structures and elements.
 - ◆ An algebraic notation is used to represent the data structure.
 - ◆ The symbols used are:
 - ◆ Equal sign, meaning “consists of”.
 - ◆ Plus sign, meaning “and”.
 - ◆ Braces $\{ \}$ meaning repetitive elements, a repeating element or group of elements.
 - ◆ Brackets $[]$ for an either/or situation.
 - ◆ The elements listed inside are mutually exclusive.
 - ◆ Parentheses $()$ for an optional element.

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Structures

- ◆ Repeating Groups

- ◆ A repeating group may be:

- ◆ A sub-form.

- ◆ A screen or form table.

- ◆ A program table, matrix, or array.

- ◆ There may be one repeating element or several within the group.

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Structures

- ◆ The repeating group may have:

- ◆ Conditions.

- ◆ A fixed number of repetitions.

- ◆ Upper and lower limits for the number of repetitions.

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Structures

- ◆ Data structures may be either logical or physical.

- ◆ Logical data structures indicate the composition of the data familiar to the user.

- ◆ Physical Data Structures Include

- ◆ elements and information necessary to implement the system

- ◆ Key fields used to locate records.

- ◆ Codes to indicate record status.

- ◆ Codes to identify records when multiple record types exist on a single file

- ◆ A count of repeating group entries.

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Structures

- ◆ Example

```
Customer Order = Customer Number +  
                  Customer Name +  
                  Address +  
                  Telephone +  
                  Catalog Number +  
                  Order Date +  
                  {Order Items} +  
                  Merchandise Total +  
                  (Tax) +  
                  Shipping and Handling +  
                  Order Total +  
                  Method of Payment +  
                  (Credit Card Type) +  
                  (Credit Card Number) +  
                  (Expiration Date)
```

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Structures

- ◆ Structural Records

- ◆ A structure may consist of elements or smaller structural records.

- ◆ These are a group of fields, such as:

- ◆ Customer Name, Address, Telephone.

- ◆ Each of these must be further defined until only elements remain.

Data dictionary

- ◇ Data Dictionary Contents
 - ◇ Data Structures
 - ◇ Structural Records
 - ◇ Example

```
Customer Name = First Name +  
                (Middle Initial) +  
                Last Name  
  
Address =       Street +  
                (Apartment) +  
                City +  
                State +  
                Zip +  
                (Zip Expansion) +  
                (Country)  
  
Telephone =     Area code +  
                Local number
```

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Elements

- ◆ Data elements should be **defined** with **descriptive information**, **length** and **type** of data information, **validation criteria**, and **default values**.

- ◆ Each element should be defined once in the data dictionary

- ◆ **Attributes** of each element are:

- ◆ Element **ID**: This is the entry that allows the analyst to build automated data dictionary entries.

- ◆ The **name** of the element, descriptive and unique

- ◆ It should be what the element is **commonly** called in most programs or by the major user of the element.

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Elements

- ◆ **Attributes** of each element are:

- ◆ **Aliases**, which are synonyms or other names for the element

- ◆ These are names used by different users within different systems

- ◆ Example, a Customer Number may be called a

- ◆ Receivable Account Number or Client Number.

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Elements

- ◆ **Attributes** of each element are:

- ◆ A short **description** of the element

- ◆ Whether the element is base or derived

- ◆ A base element is one that has been initially keyed into the system

- ◆ A derived element is one that is created by a process, usually as the result of a calculation or some logic.

- ◆ The **length** of an element

- ◆ example

Element	Length	Percent of data that will fit within the length
Last Name	11	98%
First Name	18	95%
Company Name	20	95%
Street	18	90%
City	17	99%

Data dictionary

◆ Data Dictionary Contents

◆ Elements

◆ Example

Name	Customer Number
Alias	Client Number
Alias	Receivable Account Number
Description	Uniquely identifies a customer that has made any business transaction within the last five years.
Length	6
Input Format	9(6)
Output Format	9(6)
Default Value	
Continuous/Discrete	Continuous
Type	Numeric
Base or Derived	Derived
Upper Limit	<999999
Lower Limit	>18
Discrete	Value/Meaning
Comments	The customer number must pass a modulus-11 check-digit test.

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Stores

- ◆ Data stores contain a minimal of all base elements as well as many derived elements.

- ◆ Data stores are **created** for each different data **entity**; that is, each different person, place, or thing **being stored**.

- ◆ Data Store Definition :

- ◆ The Data Store **ID**,

- ◆ The Data Store **Name**, descriptive and unique

- ◆ An **Alias** for the file,

- ◆ A short **description** of the data store

- ◆ The **file type**, either manual or computerized

Data dictionary

- ◆ Data Dictionary Contents

- ◆ Data Stores

- ◆ example

ID	D1
Name	Customer Master
Alias	Client Master
Description	Contains a record for each customer
File Type	Computer
File Format	Database
Record Size	200
Maximum Records	45,000
Average Records	42,000
Percent Growth/Year	6%

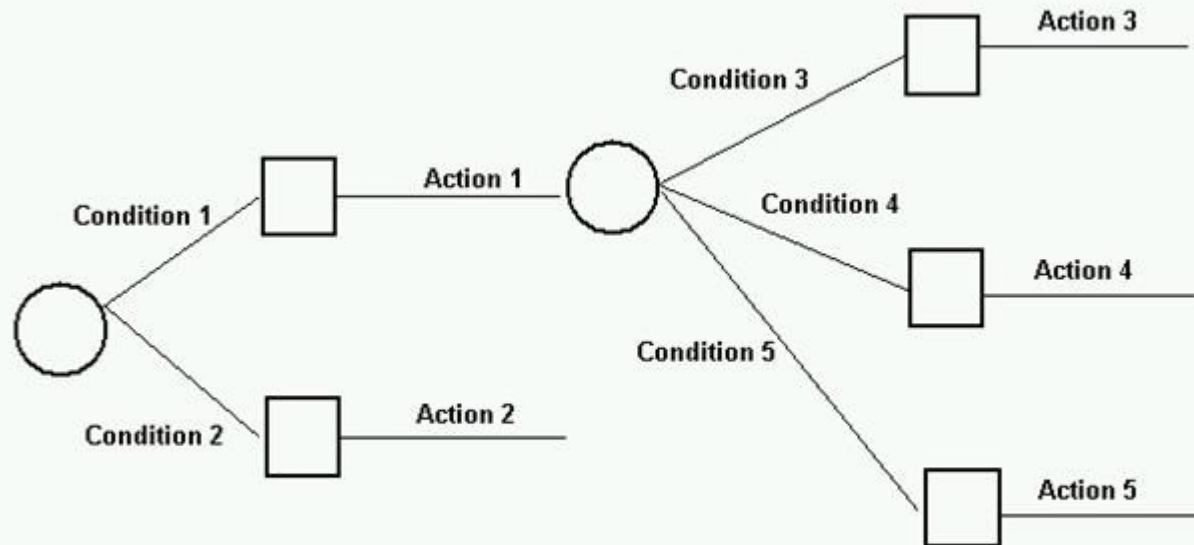
Data Set/Table Name	Customer
Copy Member	Custmast
Data Structure	Customer Record
Primary Key	Customer Number
Secondary Keys	Customer Name, Telephone, Zip Code
Comments	The Customer Master file records are copied to a history file and purged if the customer has not purchased an item within the past five years. A customer may be retained even if he or she has not made a purchase by requesting a catalog.

Decision tree

- ◇ Is a method for defining complex relationships by describing **decisions** and avoiding the problems in communication.
- ◇ It is a **diagram** that shows **alternative actions and conditions** within horizontal tree framework.
 - ◇ Thus, it depicts which conditions to consider first, second, and so on.
- ◇ Decision trees depict the relationship of each condition and their permissible actions.
 - ◇ A **square** node indicates an **action**
 - ◇ A **circle** indicates a **condition**.
- ◇ It forces analysts to consider the sequence of decisions and identifies the actual decision that must be made.

Decision tree

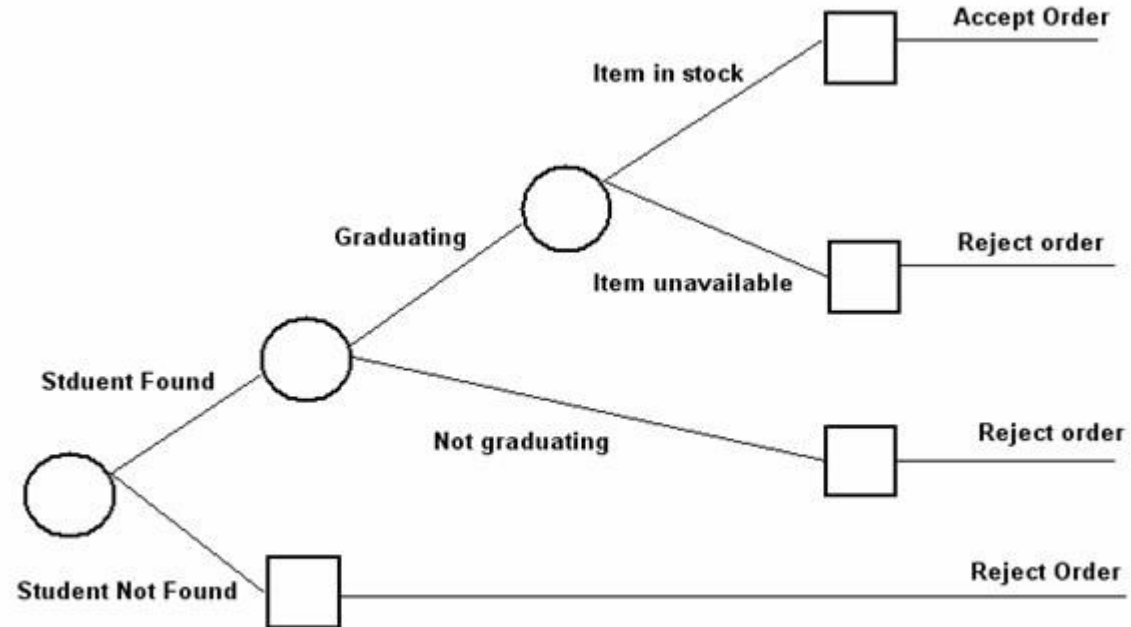
- Decision Tree Elements:



Decision tree

◇ Example

◇ Assume you have An Student Gown ordering system



Decision tree

- The major limitation of a DT is that it **lacks information** in its format to describe what other combinations of conditions you can take for testing.
- It is a single representation of the relationships between **conditions** and **actions**.
 - For example:



Exercise

- ◇ Assume a library system and draw a decision tree for it
- ◇ The library has the following rules/Policy
 - ◇ A Teacher can take a Book for full Semester
 - ◇ A Student can take a Book for two weeks
 - ◇ A Visitor can take a Book for 1 week after deposit a security Amount
 - ◇ If Book is not available then a teacher/student can reserve a book
 - ◇ If there is a delay a student is fined 50birr
 - ◇ If there is a delay a visitor is fined 100birr
 - ◇ The visitor's security is refunded after returning the book
 - ◇ Delay and fine rule doesn't apply for teachers

Decision Tables

- ◇ A method of describing the complex logical relationship in a precise manner which is easily understandable.
- ◇ It is **useful** in situations where the **resulting actions depend** on the occurrence of one or several combinations of independent **conditions**.
- ◇ It is a matrix containing **row** or **columns** for defining a **problem** and the **actions**.

Decision Table

- ◆ **Components of a Decision Table**

- ◆ **Condition Stub**

- ◆ It is in the **upper left** quadrant which lists all the **condition** to be checked.

- ◆ **Action Stub**

- ◆ It is in the **lower left quadrant** which outlines all the **action** to be carried out to meet such condition.

- ◆ **Condition Entry**

- ◆ It is in **upper right quadrant** which provides **answers to** questions asked in **condition stub** quadrant.

- ◆ **Action Entry**

- ◆ It is in **lower right quadrant** which indicates the appropriate **action** resulting from the **answers** to the conditions in the condition entry quadrant.

Decision Table

- ◆ The entries in decision table are given by decision rules
 - ◆ which define the relationships between combinations of conditions and courses of action.
- ◆ In rules section
 - ◆ Y shows the existence of a condition.
 - ◆ N represents the condition, which is not satisfied.
 - ◆ A blank - against action states it is to be ignored.
 - ◆ X (or a check mark will do) against action states it is to be carried out.

Decision Table

Example:

CONDITIONS	Rule 1	Rule 2	Rule 3	Rule 4
Advance payment made	Y	N	N	N
Purchase amount = Rs 10,000/-	-	Y	Y	N
Regular Customer	-	Y	N	-
ACTIONS				
Give 5% discount	X	X	-	-
Give no discount	-	-	X	X

Exercise

- ◇ Assume a library system and draw a decision table for it
- ◇ The library has the following rules/Policy
 - ◇ A Teacher can take a Book for full Semester
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 - ◇ If there is a delay a student is fined 50birr
 - ◇ If there is a delay a visitor is fined 100birr
 - ◇ The visitor's security amount is refunded after returning the book
 - ◇ Delay and fine rule doesn't apply for teachers

Condition s stub	Condition Entry									
Teacher	Y	Y	Y	N	N	N	N	N	N	N
Student	N	N	N	Y	Y	Y	Y	N	N	N
Visitor	N	N	N	N	N	N	N	Y	Y	Y
Issue book	Y	N	N	Y	Y	N	N	Y	N	N
Available	Y	N	-	Y	N	-	-	Y	-	-
Return	N	N	Y	N	N	Y	Y	N	Y	Y
Delay	-	-	-	-	-	N	Y	-	Y	N
Action stub	Action Entry									
1 week								X		
2 Week				X						
Full Semester	X									
Reserve a book		X			X					
Pay 50birr refine							X			
Pay 100birr refine									X	
Depose security								X		
Refund security									X	X
Return book			X			X	X		X	X

- Possible solution for the previous scenario

Go To



Chapter Three



Chapter Two



Chapter One